Tide retrospective:
closing the anharmonic partition's differentiability sorry

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Abstract

This tide closes the third and final regularity field of the anharmonic-plus-linear-perturbation Gibbs instance[1] for the 1D anharmonic potential $L(x) = \frac{\lambda}{2}x^2 + \frac{\alpha}{6}x^3 + \frac{\gamma}{24}x^4$ under the strict-coercivity discriminant $\alpha^2 < 3\lambda\gamma$, with linear perturbation $A(x) = x$. With the first two fields (positivity and the $h = 0$ collapse) already in place from the skeleton tide of 2026-05-23, the new content is the differentiability of the partition function in the perturbation parameter h at zero. The proof routes through Mathlib's differentiation-under-the-integral lemma[2] with a local dominator of shape $t\,e^{t/(2c)} \cdot |x|\,e^{-(t/2)L(x)}$ on $|h| < 1$. The instance unlocks the cross-susceptibility derivative theorem[3] in the anharmonic case, completing the formal anchor for the FDT-identity empirical validation. Roughly 160 lines added, no sorries, a single GPT-5.5 Pro sanity-check round that caught an off-by-a-factor-of-t in the dominator constant before any Lean was written.

1 Setting

The Susceptibility Primer's Threepoint arc identifies a cumulant-of-three formula $\partial_h \mathrm{Cov}_h(\phi, B)\big|_{h=0} = -t\,\kappa_3(\phi, A, B)$ through a fluctuation–dissipation argument. The argument runs through a regularity typeclass[4] which packages three analytic prerequisites for the perturbed Gibbs measure: strict positivity of the unperturbed partition, the $h = 0$ collapse of the perturbed partition to the unperturbed integral, and differentiability of the partition in h at zero with the integral-of-the-pointwise-derivative formula as its derivative. The harmonic instance — where $L = (\lambda/2)x^2$ and the partition admits a closed form by square completion — has been formalised since the `2026-05-06-tide-harmonic-gibbsregularity` run; the anharmonic mirror is the natural next step, and the FDT-identity tide-experiment arc (closed empirically at `2026-05-23-experiment-fdt-identity-1d`) requires this instance as its formal anchor.

The skeleton tide of 2026-05-23 landed the first two fields and added two public witnesses[5] covering the unperturbed Gaussian-dominated integrability and its first-moment counterpart. The third field was left as a `sorry` with a detailed proof sketch that named the strategy (differentiation under the integral with a coercivity-based dominator) but did not work out the dominator constant. This tide is the analytic-regularity follow-up promised by that sketch.

[1] Theorem `Threepoint.anharmonic_id_gibbsRegularity`; field `partition_hasDerivAt`.

[2] `hasDerivAt_integral_of_dominated_loc_of_deriv_le` in `Mathlib/Analysis/Calculus/ParametricIntegral.lean`.

[3] `Threepoint.gibbsCov_deriv_eq_neg_t_kappa3`.

[4] `Threepoint.GibbsRegularity`.

[5] The $n = 0$ and $n = 1$ cases of the dominating-Gaussian argument: `integrable_exp_neg_t_anharmonic` and `integrable_x_mul_exp_neg_t_anharmonic`.

The candidate was picked from item 5 of the v50 tide-candidates survey,[6] with the deliberation transcript in the tide-log markdown alongside this retrospective.

2 The thing formalised

For the 1D anharmonic potential $L_{\mathrm{anh}}(x) = \frac{\lambda}{2}x^2 + \frac{\alpha}{6}x^3 + \frac{\gamma}{24}x^4$ *with* $\lambda, \gamma, t > 0$ *and* $\alpha^2 < 3\lambda\gamma$*, the map*

$$h \;\mapsto\; \int_{\mathbb{R}} \exp\big(-(t(L_{\mathrm{anh}}(x) + h \cdot x))\big)\, dx$$

is differentiable at $h = 0$ *with derivative* $\int_{\mathbb{R}} (-t \cdot x)\, \exp(-(t \cdot L_{\mathrm{anh}}(x)))\, dx$*.*

The Lean signature of the instance (the `partition_hasDerivAt` field is the load-bearing part of this tide):

```
theorem _root_.Threepoint.anharmonic_id_gibbsRegularity
    {lam alpha gamma t : ℝ}
    (hlam : 0 < lam) (hgamma : 0 < gamma)
    (hdisc : alpha ^ 2 < 3 * lam * gamma)
    (ht : 0 < t) :
    Threepoint.GibbsRegularity
      (volume : Measure ℝ)
      (anharmonicPotential lam alpha gamma)
      (fun x : ℝ => x) t
```

Locally the integrand $e^{-t(L_{\mathrm{anh}}(x)+hx)}$ is, for fixed x, an entire function of h; the substance of the statement is that one can differentiate under the integral. By parity, the resulting derivative at $h = 0$ is the *first moment* of x against the unperturbed Gibbs measure scaled by $-t$, and is integrable by the public witness already in the file.

3 Proof strategy

The proof is a single application of Mathlib's differentiation-under-the-integral lemma (already cited above), with the differentiation taking place at $x_0 = 0$ on the metric ball $B(0,1) \subset \mathbb{R}$. The lemma asks for six pieces: strong measurability of F eventually near x_0, integrability of $F(x_0, \cdot)$, strong measurability of $F'(x_0, \cdot)$, a pointwise bound on $\|F'(h,x)\|$ uniform in $h \in B(0,1)$ by an integrable function of x, and pointwise differentiability of $F(\cdot, x)$ at every $h \in B(0,1)$.

The measurability pieces are immediate from continuity in x for each fixed h, and `fun_prop` dispatches them in one step after a single `unfold anharmonicPotential`. The integrability of $F(0,\cdot) = e^{-tL_{\mathrm{anh}}}$ is the public witness `integrable_exp_neg_t_anharmonic` (after collapsing the $+0 \cdot x$ by `simp only [zero_mul, add_zero]`); the strong measurability of $F'(0,\cdot) = -tx \cdot e^{-tL_{\mathrm{anh}}}$ is again `fun_prop`. The pointwise derivative is mechanical: the function $h \mapsto e^{-t(L+hx)}$ is a composition of `hasDerivAt_id`, `HasDerivAt.mul_const`, `HasDerivAt.const_add`, `HasDerivAt.const_mul`, `HasDerivAt.neg`, and `HasDerivAt.exp`. The derivative value comes out as the exponential times $-tx$; a `convert ...using 1; ring` aligns the placement of the constant factor with the expected shape.

The interesting piece is the dominator. We need a single function $D : \mathbb{R} \to \mathbb{R}$, integrable, such that $\|F'(h,x)\| \le D(x)$ for every h with $|h| < 1$. Starting from $|F'(h,x)| = t\,|x|\, e^{-t(L(x)+hx)}$

[6] `projects/automation/log/2026-05-24-v50-tide-candidates-survey.md`.

and using $|hx| \le |x|$ on the ball, we get $|F'(h,x)| \le t\,|x|\,e^{-tL(x)+t|x|}$. The natural choice of dominator combines coercivity $cx^2 \le L(x)$ (witnessed by `anharmonic_coercive`) with an AM-GM-style absorption of the linear $t|x|$ excess into the Gaussian quadratic. Specifically $t|x| \le (tc/2)x^2 + t/(2c)$ (which is the AM-GM inequality $(\sqrt{tc/2}\,|x| - \sqrt{t/(2c)})^2 \ge 0$ expanded), and the quadratic half is absorbed by half of the coercive bound: $(tc/2)\,x^2 \le (t/2)\,L(x)$. Stacking gives

$$\|F'(h,x)\| \;\le\; t \cdot e^{t/(2c)} \cdot |x| \cdot e^{-(t/2)L_{\mathrm{anh}}(x)}.$$

This is a centred polynomial-times-Gaussian-weighted-by-the-half- temperature potential. Its integrability is the same public witness as the $n = 1$ Gaussian-domination argument, applied at temperature $t/2 > 0$: `integrable_x_mul_exp_neg_t_anharmonic` takes care of $\int x \cdot e^{-(t/2)L_{\mathrm{anh}}}$, and `.norm` bridges $|x \cdot e^{\cdots}|$ to $|x| \cdot e^{\cdots}$ (since $\exp > 0$).

4 Roadblocks and resolutions

The deliberation step caught a single off-by-factor-of-t error before any Lean code was written. The first dominator draft had the absorbing constant as $e^{t^2/(2c)}$; GPT-5.5 Pro flagged this and gave the corrected derivation:

> **GPT-5.5 Pro:** For $u \ge 0$, $-\frac{tc}{2}u^2 + tu = -\frac{tc}{2}(u - 1/c)^2 + t/(2c)$, so the maximum is attained at $u = 1/c$, with value $t/(2c)$. Hence $|F'(h,x)| \le t\,e^{t/(2c)}\,|x|\,e^{-(tc/2)x^2}$.

The fix is one symbolic substitution but trying it in Lean first would have wasted at least one rebuild cycle on a downstream inequality that wouldn't close.

In Lean the proof landed in two phases. Three private helpers crystallised the analytic content: the pointwise derivative, the AM-GM step, the pointwise bound, and the bound's integrability. Then the main theorem became a structural one-shot application of the Mathlib lemma. Three friction points are worth recording.

First, `HasDerivAt.mul_const` composed with `hasDerivAt_id` gives $\mathrm{HasDerivAt}(\mathrm{id} \cdot c)(1 \cdot c)$, not $\mathrm{HasDerivAt}(\mathrm{id} \cdot c)c$: the $1 \cdot c$ does not reduce automatically in this elaboration context. The fix is a single `simpa`, but the symptom is a type-mismatch that doesn't mention the missing `one_mul` rewrite. Worth landing in `CLAUDE.md`.

Second, `field_simp` followed by `ring` on an equation of the shape $2c \cdot (\frac{tc}{2}x^2 + \frac{t}{2c}) = tc^2x^2 + t$ now *closes* the goal on its own; the trailing `ring` then errors with 'No goals to be solved'. The fresher Mathlib cache pulled by `tide-worktree create` (cf. the `tide-worktree Mathlib cache drift` memory) is strict about this idiom whereas earlier caches tolerated it. The fix is to drop the redundant `ring`.

Third, the differentiation-under-the-integral lemma is *not* in the `MeasureTheory` namespace, despite its content being measure-theoretic. It lives at the root scope after the file's `public section` declaration. Calling it with the `MeasureTheory.` prefix produces an 'unknown identifier' error; calling it bare succeeds.

5 What was Mathlib, what was new

Mathlib provided the differentiation-under-integral lemma, the standard `HasDerivAt` combinators, the `Integrable` / `AEStronglyMeasurable` algebra (with `.norm`, `.const_mul`, `.congr`), the `Continuous.aestronglyMeasurable` bridge, `fun_prop`, and the linarith / nlinarith arithmetic engines.

The new content of this tide is four private helpers[7] plus the structural assembly of the main theorem. The AM-GM helper is the only piece with any real mathematical content; the rest is bookkeeping over already-formalised seabed witnesses. Of the ~ 160 added lines, the helper proofs account for roughly two-thirds and the main theorem assembly for one-third.

6 Lessons

- For `CLAUDE.md` of the laplace repo: record the `(hasDerivAt_id h).mul_const c` returning $1 \cdot c$ rather than c gotcha and the `simpa` fix.
- For `CLAUDE.md`: record that the differentiation-under-the-integral lemma is unprefixed (no `MeasureTheory.` namespace).
- For `CLAUDE.md`: record the `field_simp; ring` pitfall on bilinear rational identities in the current Mathlib cache — `field_simp` now closes these on its own.
- For the tide skill: a single deliberation round (one query out, one response back) was enough here because the candidate was already pinned by the existing sorry's sketch. The deliberation nonetheless paid for itself by catching the dominator-constant error before the first Lean rebuild. Sanity-checking dominator arithmetic against GPT is cheap and load-bearing.
- For the `lean-formalisation` skill: separating the pointwise bound and the integrability into independent private helpers, with the main theorem as a thin assembly, kept the error surface small. Each rebuild cycle pinpointed exactly one helper.
- For future anharmonic-side tides: the dominator $t\, e^{t/(2c)}|x|\, e^{-(t/2)L_{\mathrm{anh}}}$ is the correct shape for differentiating once in h. For higher h-derivatives or higher moments ($\int x^n \cdot e^{-(t/2)L_{\mathrm{anh}}}$) the same pattern works with `integrable_pow_mul_exp_neg_anharmonic` at half temperature in place of the $n = 1$ special case.

7 Follow-ups

- **Consume the instance.** Now that all three regularity fields are sorry-free for the anharmonic-plus-linear-perturbation case, the FDT and cross-susceptibility derivative theorems[8] apply unconditionally for this potential. A short capstone tide that instantiates them and writes out the headline identities is the natural next step.
- **Higher-derivative version.** The same proof structure applies to higher h-derivatives, modulo replacing the first-moment integrability witness with the $n = 2, 3$ cases. Would be needed for direct formalisation of the flow-equation (fourth-cumulant) identity.
- **Upstream candidate.** The `amgm_t_abs_x` helper (a scalar AM-GM inequality of the form $t|x| \le \frac{tc}{2}x^2 + \frac{t}{2c}$) is fully generic and might be useful in any context where one absorbs a linear excess into a Gaussian. Worth keeping in mind for promotion to `lean-common` if a second use surfaces.
- **Anharmonic observable hypothesis.** The Threepoint module separately requires a per-observable differentiability witness; the anharmonic analogue for $\phi(x) = x^2$ (the cross-susceptibility's natural observable) is a parallel tide whose proof structure mirrors this one.

[7] `anharmonic_perturbed_pointwise_hasDerivAt`, `amgm_t_abs_x`, `anharmonic_perturbed_pointwise_bound`, and `anharmonic_bound_integrable`.

[8] The first-cumulant FDT and the cross-susceptibility identity from Threepoint.CrossSusceptibility.

Acknowledgements

One sanity-check round with GPT-5.5 Pro caught the dominator-constant error before the first Lean rebuild, and is preserved verbatim alongside the tide-log markdown. The unperturbed and first-moment Gaussian-domination witnesses were landed in the anharmonic-partition-deriv-partial tide, and the skeleton's two-of-three regularity fields in the gibbsregularity skeleton tide.